

GABRIEL ADAMS

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ACCOLADES AND CERTIFICATIONS

CyberCorps: Scholarship for Service | U.S. Office of Personnel Management Cookeville, TN

SY0-701 - CompTIA Security+ Issued: Nov. 1, 2024
Valid Through: Apr. 13, 2028 (Renewed)

CS0-003 - CompTIA CySA+ Issued: Apr. 13, 2025
Valid Through: Apr. 13, 2028

ISE - IoT Security Essentials Issued: Oct. 24, 2025
Valid Through: Oct. 23, 2028

EXPERIENCE

Vulnerability Sciences Group Intern May 2026 – July 2026
Oak Ridge National Laboratory Oak Ridge, TN

- Worked as part of a team which carried out security research on multiple devices using various vulnerability research methods such as reverse engineering using Ghidra, source code analysis, and digital signal analysis.
- Presented my work in the form of a technical presentation to an advisor committee as part of an effort to display the work that is being done by professionals early in their career at ORNL.
- Created multiple technical reports over the devices under assessment which helped to enrich the knowledge base of the team to use in future assessments.

Contract Penetration Tester Sept. 2025 – Nov. 2025
ZOE International (Nonprofit) Remote

- Conducted a cloud application penetration test under NDA to assess and strengthen security posture.
- Delivered findings and remediation recommendations to support safeguarding sensitive systems.

X163 Energy and Control Systems Security Intern May 2025 – Aug. 2025
Oak Ridge National Laboratory Oak Ridge, TN

- Performed vulnerability research and assessments on embedded devices across two projects, producing reports and presentations.
- Developed custom Rust-based fuzzing frameworks targeting RTOS firmware to uncover flaws.
- Used Ghidra to reverse engineer firmware and identify authentication bypass vulnerabilities in industrial control systems.
- Authored technical documentation and presented findings to engineering teams and DOE mentors.

Satellite Security Lead Researcher Aug. 2024 – Present
Cybersecurity, Education, Research, and Outreach Center (CEROC) @ TN Tech Cookeville, TN

- Built digital twin testbeds to analyze satellite communication security.
- Developed data analysis tools for transmission data patterns and timing anomalies.
- Researched side-channel countermeasures for satellite cryptographic modules.

Embedded Systems Security Analyst May 2024 – Aug. 2024
Center for Energy Systems Research @ TN Tech Cookeville, TN

- Secured SCADA-to-device communication infrastructure and managed RTAC configurations.
- Developed portable, secure RTAC images compatible with HIL simulation systems.
- Helped complete a multi-million-dollar CESR energy systems research project.

IT Security Intern

Rebo Lighting and Electronics

May 2024 – Aug. 2024

Sparta, TN

- Implemented SIEM systems for network traffic monitoring and alert correlation.
- Conducted vulnerability assessments and managed mitigation plans.
- Supported ISO 27001 certification through policy creation, tool deployment, and staff training.

AI Drone Security Researcher

Cybersecurity, Education, Research, and Outreach Center (CEROC) @ TN Tech

July 2023 – May 2024

Cookeville, TN

- Created a multimodal AI model to detect false-data injection, DoS, and MITM attacks from flight logs.

Embedded Systems Engineer

Center for Energy Systems Research @ TN Tech

Jan. 2023 – May 2024

Cookeville, TN

- Developed embedded C software for wireless energy transfer systems.
- Collaborated on research leading to a university publication credit as an engineer.

EDUCATION

Tennessee Technological University

B.S. in Computer Science, Cybersecurity Concentration

Cookeville, TN

Aug. 2022 – Dec. 2025

Tennessee Technological University

M.S. in Information Assurance and Cybersecurity

Cookeville, TN

Jan. 2026 – Dec. 2026

LEADERSHIP AND EXTRACURRICULAR

Defensive Cybersecurity Interest Group Lead | *Jan. 2024 – Present*

Tennessee Tech

CyberEagles Vice-President | *Aug. 2024 – Present*

Tennessee Tech

TECHNICAL SKILLS

Programming & Development: C, C++, Python, Rust, Embedded C, x86/ARM Assembly, Bash/Shell.

Embedded & ICS: Firmware reverse engineering, side-channel analysis, protocol fuzzing, Power System Protection, UART, SPI, I2C, SWIM, JTAG.

Security Operations: SIEM, Threat Hunting, Vulnerability Assessment, Incident Response, Forensics, Log Analysis.

Networking & Virtualization: Firewalls, IDS/IPS, DNS, Linux, Windows, Proxmox, Docker, Cloud Security.

Compliance & Cryptography: PKI, IAM, Risk Management, ISO 27001, SSDLC, Encryption.

IoT Security: Wireless Protocol Security, Android Application security, BLE Security, RF Transmission protocols, OTA Update Security, Hardware Root of Trust, penetration testing.

COMPETITIONS

MITRE Embedded Capture The Flag (eCTF) | *Aug. 2025 – May 2026*

Boston, MA

- Served as technical lead for my team during both the development and attack phases of the competition.

Cybersecurity Collegiate Defensive Competition (CCDC) | *Jan. 2025 – May 2025*

- Led the university's first-ever CCDC team to a 1st-place regional finish as captain.

Cyber Resiliency and Measurement Challenge (CRAM) | *Aug. 2024 – Oct. 2024*

Dahlgren, VA

- Led a team of five in a NAVSEA competition to apply LLMs, ML, and data science for system resilience analysis.

HOBBIES

IoT Pentesting | *Jan. 2022 – Present*

- * Carry out security assessments on consumer IoT devices to find vulnerabilities to then coordinate with the manufacturer, leading to multiple confirmed vulnerabilities and a bug bounty on the HakcerOne platform.
- * Developed open-source tools to automate firmware analysis and vulnerability discovery.